LAGUNA STATE POLYTECHNIC UNIVERSITY

Bachelor of Science in Information Technology

Enterprise Server Deployment and Operating System Installation

Ubuntu Server Installation Guide

A repeatable, evidence-based installation guide for the Week 3 ITEP 414 portfolio project.

Project Information	Details
Course	ITEP 414 - System Administration and Maintenance
Program	Bachelor of Science in Information Technology
Student	[Your Full Name]
Lab platform	[Oracle VirtualBox / VMware Workstation]
Ubuntu ISO	Ubuntu Server 26.04.1 LTS, unless your instructor provides a different approved ISO
Document status	Template - complete with your actual lab results

Prepared for ITEP 414 - System Administration and Maintenance

Working template: replace bracketed fields and add genuine lab evidence before submission.

1. Project Overview

This project documents the deployment of a clean Ubuntu Server virtual machine for a hypothetical startup environment. The server is intended as a foundation for future file sharing, remote administration, database hosting, web hosting, and internal services.

The guide covers installation, initial configuration, verification, BIOS-versus-UEFI concepts, the boot process, and a summary of the Windows Server evaluation activity. It is written so another administrator can reproduce the procedure.

Learning Objectives

- Explain the purpose of an operating system in an enterprise environment.
- Differentiate BIOS and UEFI and describe the boot process.
- Install Ubuntu Server in a virtual machine and configure core settings.
- Enable SSH for remote administration and verify the service.
- Record results and produce reproducible technical documentation.

Important Evidence Rule

Do not mark a test as passed until you have run it and captured your own evidence. The result fields in this guide are intentionally left blank for your lab session.

2. Software and Preparation

Requirement	Selection / Notes
Virtualization software	Oracle VirtualBox or VMware Workstation
Ubuntu ISO	Use the latest supported LTS ISO or the ISO specified by the instructor. As of 26 October 2026, Ubuntu 26.04.1 LTS is the current LTS point release listed by Ubuntu.
Windows Server ISO	Windows Server 2025 Evaluation ISO from the official Microsoft Evaluation Center, for the bring-home activity.
Host resources	Ensure the host has enough free RAM, CPU capacity, and disk space for the virtual machines.
Evidence folder	Create Week03/screenshots/ and save readable, unaltered screenshots of your own results.

Safety and Preparation

- Download ISO files from official vendor sites and verify the file name and version.
- Use a new virtual disk for the lab. The guided entire-disk option erases the selected target disk; never select a host disk containing personal data.
- Use a strong, unique password. Do not place passwords, recovery codes, or private SSH keys in GitHub.
- Record the actual VM firmware mode, virtual disk layout, filesystem, IP address, and installed ISO version.

3. Virtual Machine Specification

Component	Required setting
VM name	Ubuntu-Server-Week03
Memory	4 GB RAM
CPU	2 virtual processors
Storage	40 GB virtual disk (VDI or VMDK)
Network	NAT, or Bridged only if instructed
Optical drive	Attach the Ubuntu Server ISO
Firmware	Use the instructor-required/default VM setting and record whether BIOS or UEFI is enabled.

4. Ubuntu Server Installation Procedure

1. Open VirtualBox or VMware and create a new virtual machine named Ubuntu-Server-Week03.

2. Select a 64-bit Linux/Ubuntu guest type where the platform asks for an operating system family.

3. Assign 4 GB RAM and 2 virtual CPUs. Create a new 40 GB virtual disk in VDI or VMDK format.

4. Set the network adapter to NAT unless your instructor requires Bridged networking.

5. Attach the Ubuntu Server LTS ISO to the virtual optical drive, then start the VM.

6. At the installer, select English and choose the appropriate keyboard layout, such as English (US).

7. For networking, accept DHCP unless instructed otherwise. Record any address shown by the installer; verify it again after the first login.

8. Leave proxy and mirror settings at their defaults unless your environment requires a proxy or a different mirror.

9. For storage, choose Guided - Use Entire Disk and select only the 40 GB virtual disk created for this VM. Review the proposed layout before confirming.

10. Create a non-root administrative account using your own name, a suitable username, and a strong password. Set the hostname to server01.

11. On the SSH screen, select Install OpenSSH Server. Do not add optional packages unless your instructor instructs you to.

12. Review the summary and confirm installation. Wait for the installer to finish, then reboot.

13. If prompted, detach/eject the installation ISO so the VM boots from its virtual disk. Log in with the account you created.

Record the Actual Storage Layout

Field	Actual value from your installer / VM
Virtual disk size	[e.g., 40 GB - confirm actual value]
Partition table / scheme	[Record what the installer created; do not assume]
Filesystem(s)	[Record actual filesystem(s), e.g., ext4 if shown]
LVM used?	[Yes / No / Not sure - verify]
Firmware mode	[BIOS / UEFI]

5. Initial Verification

Run each command in the Ubuntu Server console. Capture a screenshot that shows the command and its meaningful output. If the terminal output is long, adjust the terminal size so the relevant result is readable.

Task	Command	What to confirm
Login	Log in at the console	Your created user can authenticate.
Hostname	hostname	Output is server01.
IP address	ip addr	An address is assigned to the active network interface; record the correct interface and address.
Internet / DNS	ping -c 4 google.com	Replies are received. If it fails, diagnose networking and DNS before concluding.
Package index and updates	sudo apt update sudo apt upgrade -y	Commands complete without unresolved errors; record any reboot notice.
SSH service	systemctl status ssh	Service is active (running).

Useful Follow-up Commands

```
ip route
resolvectl status
sudo systemctl enable --now ssh
sudo ss -tlnp | grep ':22'
```

The follow-up commands are for diagnosis or confirmation. If SSH was not installed during setup, install it only as needed and document the change: sudo apt install openssh-server. Then run sudo systemctl enable --now ssh.

Verification Results Log

Check	Actual result	Evidence filename	Status
Login	[Fill in]	[03_login.png]	[Pass / Fail]
Hostname	[Fill in]	[04_hostname.png]	[Pass / Fail]
IP address	[Fill in]	[05_ip_addr.png]	[Pass / Fail]
Internet connectivity	[Fill in]	[06_ping_google.png]	[Pass / Fail]
System update	[Fill in]	[07_apt_update_upgrade.png]	[Pass / Fail]
SSH service	[Fill in]	[08_ssh_status.png]	[Pass / Fail]

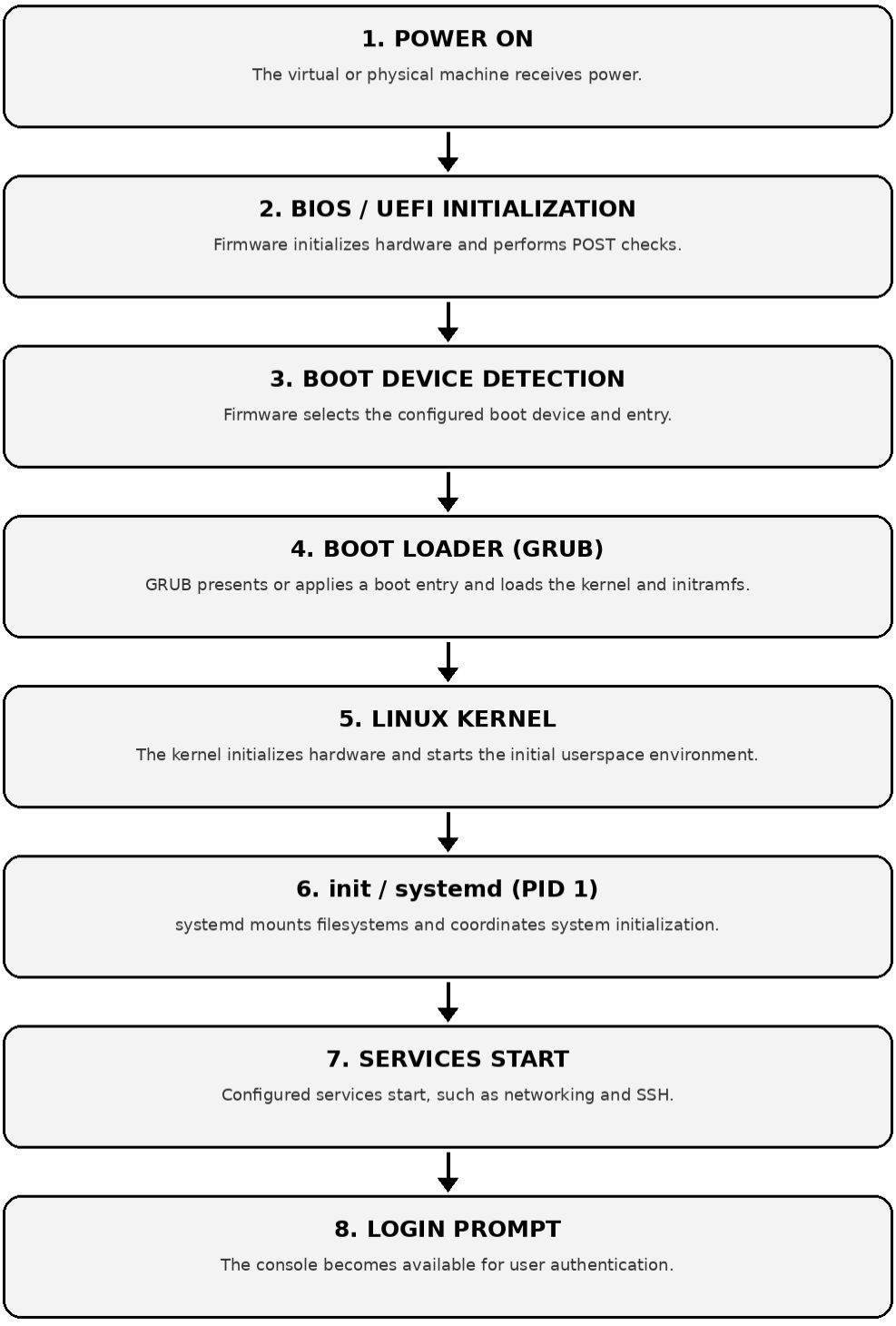
6. BIOS vs. UEFI and the Boot Process

BIOS is the legacy firmware interface commonly associated with MBR booting. UEFI is the modern firmware interface with standardized boot services and support for booting EFI applications. UEFI systems commonly use GPT and can use Secure Boot when supported and enabled. Disk capacity depends on the partitioning scheme and implementation; classic MBR layouts are commonly limited to about 2 TiB with 512-byte logical sectors.

The simplified Ubuntu boot sequence for this project is: Power On -> BIOS/UEFI Initialization -> Boot Device Detection -> GRUB -> Linux Kernel -> init/systemd -> Services Start -> Login Prompt.

Ubuntu Server Boot Process

From power-on to authenticated console login



Note: exact screens and service timing depend on firmware, VM settings, and installed services.

7. Windows Server Evaluation - Bring-Home Activity

Use the official Microsoft Evaluation Center to download the Windows Server evaluation ISO. The current listed LTSC release is Windows Server 2025. Microsoft states that the evaluation expires after 180 days; follow the current Evaluation Center instructions and your instructor's requirements.

Step	Action / setting
1	Create a separate VM named Windows-Server-Week03.
2	Suggested lab baseline: 2 vCPUs, 4 GB RAM, 60 GB virtual disk, and NAT networking. Adjust if the official installer or instructor requires more.
3	Attach the Windows Server 2025 Evaluation ISO and start the VM.
4	Choose the evaluation edition and, for a beginner-friendly lab, the Desktop Experience option if available and permitted.
5	Complete setup, set a computer name such as WIN-SRV-WK03, and create a strong Administrator password.
6	Log in successfully, install updates, and capture genuine screenshots of the installation and first login.
7	Record the exact edition, VM settings, hostname, date, and any issues. Never include the Administrator password in the report.

8. Troubleshooting Notes

Symptom	Checks / possible response
No IP address	Check the VM adapter is enabled and set to NAT; renew/restart networking if appropriate; inspect ip addr and ip route.
DNS or ping failure	Check ip route and resolvectl status. Test name resolution with getent hosts google.com. Some networks block ICMP, so distinguish ping failure from complete network failure.
SSH unit not found	OpenSSH Server may not have been selected. If allowed, run sudo apt install openssh-server, then sudo systemctl enable --now ssh.
Cannot access the VM remotely	Confirm the service is active and the VM network mode supports the intended access. NAT normally requires port forwarding for host-to-guest inbound connections.
Update command fails	Check connectivity, system time, repository configuration, and the full error message. Retry only after correcting the underlying issue.
VM boots back to installer	Detach/eject the ISO or change boot order so the virtual disk is first.

9. Problems Encountered and Solutions Implemented

Complete this section after the lab. Do not invent issues or claim fixes that you did not perform.

Problem observed	Evidence / error	Root cause	Solution actually applied
[Problem 1]	[Exact message]	[Cause]	[Fix and verification]
[Problem 2]	[Exact message]	[Cause]	[Fix and verification]

10. References

[1] Ubuntu Server installation documentation. https://ubuntu.com/server/docs/installation/. Accessed 26 October 2026.

[2] Ubuntu Server OpenSSH documentation. https://ubuntu.com/server/docs/how-to/security/openssh-server/. Accessed 26 October 2026.

[3] Ubuntu Server package management. https://ubuntu.com/server/docs/how-to/software/package-management/. Accessed 26 October 2026.

[4] Ubuntu current releases. https://ubuntu.com/project/docs/release-team/list-of-releases/. Accessed 26 October 2026.

[5] Ubuntu release cycle and support. https://ubuntu.com/about/release-cycle. Accessed 26 October 2026.

[6] Microsoft Windows Server 2025 Evaluation Center. https://www.microsoft.com/en-us/evalcenter/download-windows-server-2025. Accessed 26 October 2026.

[7] Microsoft Windows Server release information. https://learn.microsoft.com/en-us/windows/release-health/windows-server-release-info. Accessed 26 October 2026.

[8] Microsoft Secure Boot overview. https://learn.microsoft.com/en-us/windows-hardware/drivers/bringup/secure-boot. Accessed 26 October 2026.

[9] UEFI Forum specifications. https://uefi.org/specifications. Accessed 26 October 2026.

[10] Rocky Linux releases. https://docs.rockylinux.org/releases/. Accessed 26 October 2026.

[11] Rocky Linux DNF package manager documentation. https://docs.rockylinux.org/guides/package_management/dnf_package_manager/. Accessed 26 October 2026.

[12] Rocky Linux official site. https://rockylinux.org/. Accessed 26 October 2026.

[13] Oracle VM VirtualBox User Manual. https://www.virtualbox.org/manual/. Accessed 26 October 2026.

[14] GNU GRUB manual. https://www.gnu.org/software/grub/manual/grub/grub.html. Accessed 26 October 2026.

[15] systemd manual. https://www.freedesktop.org/software/systemd/man/latest/systemd.html. Accessed 26 October 2026.